BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: MRD-10TH 2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** VERSION: C

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A, B, C, or D.

Q1. Which of the following is an alcohol?

(A) CH_3-O-CH_3
(B) CH_3-CH_2-OH
(C) CH_3COOH
(D) CH_3-COH

Q2. Substitution reaction is the characteristic of .....

(A) Alkynes
(B) Alkenes
(C) Alkanes
(D) None of these

Q3. Catenation is the linkage of carbon with other .....

(A) Oxygen atoms
(B) Carbon atoms
(C) Nitrogen atoms
(D) Halogens

Q4. Triglycerides are the building blocks of .....

(A) Proteins
(B) Carbohydrates
(C) Lipids
(D) Vitamins

Q5. The substance which completely dissociate in an aqueous solution and give a higher concentration of OH^- ions is called .....

(A) Strong acid
(B) Strong base
(C) Weak acid
(D) Weak base

Q6. The unit of Kc for the reaction $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$ is .....

(A) $mol \cdot dm^{-3}$ (B) $mol^{-1} \cdot dm^{-3}$

(C) $mol^{-1} \cdot dm^3$ (D) $mol \cdot dm^3$

Q7. The pOH of 0.001M solution of nitric acid is .....

(A) 0.001 (B) 10.0

(C) 11 (D) 14

Q8. The gas which protects the earth surface from ultraviolet radiations is .....

(A) CO_2 (B) N_2

(C) O_3 (D) O_2

Q9. Permanent hardness of water is removed by adding .....

(A) Washing soda (B) Soda lime

(C) Caustic soda (D) Baking soda

Q10. Water gas is a mixture of .....

(A) CO and H_2 (B) C and H_2

(C) CO and H_2O (D) C and H_2O

Q11. When sodium bicarbonate is heated to get sodium carbonate, all are produced except .....

(A) CO (B) CO_2

(C) Na_2CO_3 (D) H_2O

Q12. Blister copper is refined through .....

(A) Concentration (B) Reduction

(C) Smelting (D) Electrolysis

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks.

(i) Describe Arrhenius concept of acids and bases. Give examples.

(ii) Define and explain Law of Mass Action.

(iii) Calculate the pH of 0.02 M solution of NaOH.

(iv) How alkyl radicals are formed? Give examples.

(v) Which one is more reactive between alkane and alkene? Explain.

(vi) Write down the products of sucrose hydrolysis.

(vii) Explain ozone layer depletion.

(viii) Describe why water is important for us?

(ix) Enlist the different uses of urea.

(x) Differentiate between simple lipids and complex lipids.

(xi) How alkenes are prepared by dehydration of alcohols?

Section-C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3.

(a) [5] What are salts? Write the different types of salts.

(b) [2] Distinguish between Forward and Reverse reactions.

Q4.

(a) [4] What are pollutants? Distinguish between primary and secondary pollutants.

(b) [3] Describe the causes of hardness of water.

Q5.

(a) [4] Vitamins are vital for us. Why?

(b) [3] Sketch the flow sheet diagram of Solvay process.

Q6.

(a) [4] Write the general characteristics of organic compounds.

(b) [3] Draw the Dot, Cross and Structural formula of the following. (i) Propene (ii) Butyne (iii) Pentane